The Effects of Deprecation on Deep Learning Questions and Answers on Stack Overflow

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Stack Overflow is a powerful tool for developers in solving problems and learning from others. However, it relies on the community to provide quality questions and answers. Considering that these questions may become obsolete over time as a result of deprecation, we study the effects of deprecation on the quality of questions and answers about deep learning on Stack Overflow. It is found that questions involving a deprecated API are calculated at a higher quality than expected, increasingly useful to developers after the deprecation.

CCS Concepts: • **Computing methodologies** → *Artificial Intelligence.*

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1 INTRODUCTION

Stack Overflow is a powerful tool for developers. The online Question and Answer forum allows for developers to obtain some degree of mentorship by asking questions and receiving answers. Further, developers can search previous questions to see a solution to a similar problem. With crowd-sourced developer forums, discussion reflects what is important to developers.

Python2 officially reached end of life at the beginning of the year and developers look to Guido van Rossum for the future of Python. That said, there are still companies that use Python2 and many computers still have Python2 installed. By contrast, the recent rise of micro frameworks creates a different problem, where extensions are not maintained and any solution referencing these extensions become obsolete.

Similarly, different frameworks and APIs stop being maintained and developers use them regardless. Ideally, after a deprecation or end of life, those using the framework or API transition to a supported version as a replacement. However, as evidenced by the continued use of Python2, this is clearly not the case. Furthermore, since documentation for previous versions often become difficult to find, community forums and blogs are a good way to support these developers.

Deprecation has a number of effects. Any updates to a more recent version requires modification to the code-base. Developers must ensure that their architecture is loosely coupled to protect the system from deprecation. Finally, any new developers seeking mentorship via blogs and forums, will encounter obsolete solutions that do not support the newer version of the frameworks involved. In this way, deprecation affects the experience of developers on these blogs

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and forums. In this study, we explore the effects of deprecation on the quality of questions about deep learning and their answers on Stack Overflow, a crowd-sourced developer forum.

2 WHY QUALITY OF ANSWERS?

As time progresses and a framework expands, deprecation of an API or extension is normal. However, the effects of a deprecation are clear. Since the API is no longer usable, any blog or tutorial with said API becomes obsolete. While Stack Overflow can act as a place for people to share solutions to a common problem, if the solution provided becomes obsolete over time, then following users no longer are able to use the solution provided. As a result, there raises the question of how the quality of the questions and answers provided are affected.

On the other hand, some developers could specifically ask questions about the deprecation of an API. Questions like these are only useful to people currently upgrading to a newer version of the framework. In this way, any use of the API in the developer's project need to be updated. The use-case for these questions are very specific.

Crowd-sourced developer forums have multiple different types of questions that provide value in one way or another. However, they also rely on the community to provide high quality discussion. Unlike a peer reviewed paper, the answers provided might not be general enough to help all developers that have the problem. In the case of deprecation, if an answer becomes obsolete and is not updated, then it is no longer useful to those in the future. On the other hand, the answer becomes more valuable to those who continue to use an API after it has been deprecated. Since the documentation is updated to reflect the most recent version, the answers provided become increasingly useful for these developers.

Since crowd-sourced developer forums depend on the community to ask and answer questions well, the quality of these questions and answers is important. Since deprecation provides the situations outlined, where questions are less useful to one group but increasingly useful to a different group, it is important to assess the effect of deprecation on the quality of discussion.

We focus on the case of users looking at answers to previously asked questions. Since the answer to these questions will often aid the user, we study the question and answer as a unit.

3 METHODOLOGY

To help direct the findings, the following research questions were created:

(1) What is the deprecation status of the deep learning libraries in our study?
(2) Do users discuss the API versioning for the selected frameworks explicitly?
(3) What are the Deep Learning API versioning issues that are discussed on Stack Overflow?
(4) How do Deep Learning API versioning issues affect the quality of answers in Stack Overflow?

The frameworks studied are Tensorflow, Keras, PyTorch, Tessaract, Caffe, Theano, and PySpark. These were chosen in reference to the proposal by Gonzalez, which states that future research must contain TensorFlow and Tesseract[4]. In addition, PySpark was added because of it's comprehensive documentation in reference to its deprecated APIs.

3.1 Deprecation Status

We search the deep learning framework's GitHub pages to find the the deprecations per release. Furthermore, we extend this by adding a set of keywords that are used in previous works to identify the deprecated APIs. Specifically, we use the terms found by Zhou in the related works, the prefix for deprecation, deprecated, deprecating, to search for the

target APIs from the official documents of the frameworks[20]. We include any APIs that are removed without being deprecated, as Thung *et al.* notes this edge case[14]. The results identify the APIs that we search on Stack Overflow.

3.2 Data Collection

All questions and their answers in reference to the frameworks will be collected from Stack Overflow. The data will then be filtered to contain only the deprecated APIs. Then, the questions are filtered further, so the question must be asked after the release date of the version that deprecated the libraries. In this way, all questions in the final dataset are obsolete and reference a deprecated library.

To further investigate any explicit references to deprecated libraries, we adopt a similar filter to the related work, An Empirical Study of Obsolete Answers on Stack Overflow [20]. The following keywords were selected to filter any questions referring to deprecation: *"deprecate", "deprecation", "out of date", "outdated", "obsolete"*. This dataset supports research question 2.

3.3 Determining the Issues Discussed

We investigate the issues discussed via LDA topic modeling and a manual investigation. LDA was applied for topic modeling to extract the relevant topics for both the datasets, the deprecated libraries and explicit reference. This served two purposes. First, this helps provide evidence for or against assumptions made about the topics of discussion on Stack Overflow. Second, while a question might mention deprecation, it is not necessarily primarily about deprecation. By extracting the topics of discussion, we are better able to represent the core component of a question and answer.

We manually select a sample with 95% confidence of the collected data to investigate whether the users discuss the API versioning explicitly. We note the topics discussed among question and answer.

3.4 Hypothesis Testing

We conduct two-sided hypothesis tests on the quality of a population of questions about deprecation and the quality of a population of questions that are not about deprecation. However, since the data is heavy tailed, it does not abide by the assumptions of a T-Test.

To counteract this, we conduct three different hypothesis tests to evaluate the data. For the first test, we define a threshold for the top 5% of a population and then conduct a standard t-test on the bottom 95th quantile of the data. This way, we ignore any outliers to the data. Since the data is heavy-tailed, it is approximately log-normally distributed. The second test uses this to conduct a t-test on the logarithm of the data. Finally, a non-parametric Wilcoxon Sign Rank Test is conducted. These three tests are finally compared together to determine if there is a true difference in quality. Since the Wilcoxon rank test was the only test that properly met the assumptions, the Wilcoxon test was considered more heavily than the other tests.

3.5 Regression Model

We build a regression model using a Random Forest Regression to extract feature importance in predicting quality. Since quality is calculated as a filter of other predictor variables, these variables will dominate any non-parametric regression model that uses them. To account for this, we add a maximum number of features to be used at any split during training and remove *view_count* from the predictors. While this results in a decrease in prediction accuracy, it also allows all other features to be used.

The regression model is trained on both questions about question with an API post-deprecation and questions explicitly referencing deprecation. A Boolean variable is created to state which dataset the question originates. The topics found for the question and answer via LDA are input as a predictor variable for the model. The topics found in questions about a deprecated API are labeled by IDs ranged [0,19]; topics for questions with an explicit reference to deprecation are labeled by their ID + 20, creating a label with range [20, 39].

4 EXPERIMENTS

4.1 Experimental Setup

We wish to study the effect of deprecation on the quality of answers in forums. To do this, we define two variables of interest:

(1) Quality of an Answer
(2) Discussion of Deprecation

We study the *Quality of an Answer* as a response variable to a predictor variable *Discussion of Deprecation* on a Community Q&A Developer Forum. The Community Q&A Developer Forum we will study is *Stack Overflow*, a popular tool among developers. The necessary data is collected from Stack Overflow.

4.2 Data Collection

Three different datasets were collected for this study, as outlined in the Methodology.

The Stack Exchange Query Tool was used to select 129802 questions with 100707 answers in reference to the deep learning frameworks chosen. This dataset acts as a representation of all questions about the deep learning frameworks in study. This serves as our base understanding of a question about the deep learning frameworks studied. Then questions were filtered on if they referenced the deprecated libraries and were asked after the libraries were deprecated, using the release date of the release that deprecated them, and found 7468 questions with 4967 answers.

Finally, we filtered again on explicit reference to deprecation using the technique outlined by Zhang. This yielded 2088 questions with 1525 answers. For this dataset, since all questions reference deprecation, all unanswered questions were dropped during analysis.Using this method, it was found that 2088 questions with 1525 answers explicitly referenced deprecation status. It is interesting to note there were 562 questions in which the answer explicitly referenced the deprecation status of whatever library was in focus, despite the question not asking about deprecation.

Distribution of Questions:

Containing Deprecated API

Framework	Number of Questions	Percent of Total
Caffe	94	1.26
Keras	6828	91.43
Theano	691	9.25
Tensorflow	4610	61.73
Tessaract	45	0.6
PyTorch	99	1.33
PySpark	99	1.33

Explicit Reference to Deprecation

Framework	Number of Questions	Percent of Total
Caffe	65	4.26
Keras	451	29.57
Theano	99	6.49
Tensorflow	1073	70.36
Tessaract	17	1.11
PyTorch	77	4.66
PySpark	112	7.34

Note that these percentages do not add up to 1 because many questions reference multiple libraries.

4.3 Metric of Quality

The idea of quality is subjective and depends on what an individual might find worthwhile. In the related works, there are several different metrics to define quality. Stack Overflow has a subjective measure of question and answer quality via a voting system. This has high correlation with how many views a question has. To some degree, the popularity of a question can be considered a measure of how important a question is. Mondal notes the importance of reproducibility of issues in Crowd-Sourced Developer Forums[7]. In the related works, many different metrics for quality are proposed and studied.

We adopt a similar method to Ravi *et al.* by standardizing the community score with the view count[9]. In this research, Ravi describes the quality of a question q_i as $q_i = \frac{s_i}{v_i}$, where q_i is the quality of the answer, s_i is the community score of the question, and v_i is the view count. A question with score satisfying $q_i > 0.001$ is then labeled "good".

We define the quality of a question and answer as $q_i = \frac{sq_i + sa_i}{v_i}$, where q_i is the quality of the question and answer pair, sq_i is the community score of the question, sa_i is the community score of the answer, and v_i is the view count of a question. We study the question and answer pair because an answer is inherently dependent on the question. Then we say that a larger q_i is indicative of a higher quality.

Kavaler notes that only using the view count of a question introduces bias[5]. On the other hand, as noted by Zhou1, if a question is not solved within two days, it will likely not be solved[21]. For the purposes of this study, we analyze the question and answer in tandem and will not adjust for the difference in time. Kavaler also notes an underlying assumption that as view count increases, the score will increase as viewers up-vote a question or answer. However, as time passes since the deprecation of an API, there is the underlying assumption that any question about the API or its deprecation is not needed and viewed less. This assumption counteracts the assumption that over time the view count increases.

4.4 Metric of Discussion of Deprecation

In contrast to quality, creating a metric for the discussion of deprecation is more clear. We define two metrics for the discussion of deprecation:

(1) an explicit mention of the root *deprecate*
(2) the main topic of the question is deprecation

The keywords used to determine an explicit mention of the root *deprecate* are outlined in the methodology. However, since not all questions that have these keywords will be explicitly about deprecation, it must be supplemented. We apply LDA Topic Modeling to assess if the overall topic discussed in the question and answer is deprecation. This filter excludes any secondary topics discussed.

4.5 Deprecation Status of Deep Learning Libraries

We collect the deprecated APIs from the GitHub documentation on releases for all studied frameworks. To reduce noise, any deprecated parameters were ignored. We include any APIs removed without being deprecated.

4.6 Topic Modeling and Manual Investigation

We investigate what deep learning API versioning issues are discussed via LDA Topic Modeling and a manual investigation of the data. We apply LDA to both datasets, questions about an API post deprecation and questions explicitly referencing deprecation. From this, we also determine what the overall main topic of a question is by selecting the topic that contributes most to the question. These topics are later used in regression to investigate how deep learning API versioning issues affect the quality of answers on Stack Overflow. We support this via a manual investigation of the questions about an API after it has been deprecated.

4.7 Hypothesis Testing

We conduct two-sided hypothesis tests on the quality of a population of questions about deprecation and the quality of a population of questions that are not about deprecation. However, since the data is incredibly heavy tailed, it does not abide by the assumptions of a T-Test.

To counteract this, we conduct three different hypothesis tests to evaluate the data. For the first test, we simply define a threshold for the top 5% of a population and then conduct a standard t-test on the bottom 95th quantile of the data. This way, we ignore any outliers to the data. Since the data is heavy-tailed, it is approximately log-normally distributed. The second test uses this to conduct a t-test on the logarithm of the data. Finally, a non-parametric Wilcoxon Sign Rank Test is conducted. These three tests are finally compared together to determine if there is a true difference in quality. All questions without answers were dropped for these tests.

4.8 Regression Model

We build a regression model using a Random Forest Regression to extract feature importance in predicting quality. Since quality is calculated as a filter of other predictor variables, these variables will dominate any non-parametric regression model that uses them. To account for this, we add a maximum number of features to be used at any split during training and remove *view_count* from the predictors. While this results in a decrease in prediction accuracy, it also allows all other features to be used.

The regression model is trained on both questions about question with an API post-deprecation and questions explicitly referencing deprecation. A Boolean variable is created to state which dataset the question originates. The topics found for the question and answer via LDA are input as a predictor variable for the model. The topics found in questions about a deprecated API are labeled by IDs ranged [0,19]; topics for questions with an explicit reference to deprecation are labeled by their ID + 20, creating a label with range [20, 39]. In this way, we add two predictor variables to represent versioning.

5 RESULTS

5.1 Deprecation Status of Libraries in Question

We collect the deprecated APIs from the GitHub documentation on releases for all studied frameworks. A total of 217 APIs were found. To reduce noise, any deprecated parameters were ignored. The distribution of deprecated APIs and average APIs deprecated per version are shown in the table to the side. We define a version as a release of the form: *X.X.0*.

One interesting result was both Keras and Caffe2 deprecated the entire Framework in their final release, instead deciding to work solely with Tensorflow and

PyTorch respectively. Considering that Tensorflow and PyTorch are both primarily developed by large mega-corporations, Google and Facebook respectively, the deprecation of these two frameworks provides an interesting piece of insight for the future of deprecation in Open Source Deep Learning Frameworks.

Framework	Number of APIs	Mean Deprecation/Release
Caffe	8	2.0
Keras	30	10.0
Theano	43	5.375
Tensorflow	33	6.6
Tessaract	11	2.2
PyTorch	56	5.091
PySpark	36	5.143

5.2 Differences in Quality

Parameter Mean	All Questions	Deprecated APIs	Explicit Reference
Answer Rate	0.775851	0.665104	0.730364
Question Score	0.788431	0.813203	2.270594
View Count	438.5951	441.5287	2656.1160
Answer Count	0.965720	0.987547	1.321360
Question Comment Count	1.510980	1.535351	1.589559
Answer Score	0.9453507	0.8524260	1.650492
Answer Comment Count	1.615008	1.454600	1.057049
Quality	0.006778	0.012572	0.004156

From the table we can see questions that referenced a deprecated API are less likely to be answered, but overall the questions had the highest calculated quality. On the other hand, although there were the fewest questions that explicitly referenced deprecation, on average the questions were more popular, had more answers, and had larger views but had a lower score for quality.

Quality Across the Frameworks

Deprecated API

Framework	Number of Questions	Mean Quality
Caffe	48	0.005314
Keras	1468	0.011525
PySpark	60	0.027494
PyTorch	17	0.032739
Tensorflow	3146	0.013201
Theano	202	0.006398

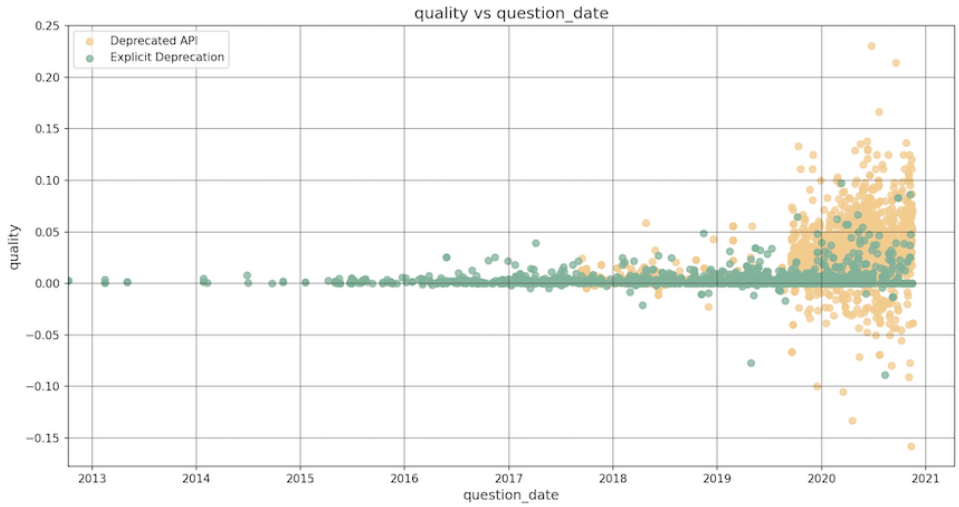
Explicit Reference to Deprecation

Framework	Number of Questions	Mean Quality
Caffe	65	0.002547
Keras	116	0.003320
PySpark	112	0.003767
PyTorch	64	0.007194
Tensorflow	1073	0.0043371
Theano	78	0.003049

It is worth noting that the general distribution of quality is the same across both populations of questions. However, for all frameworks, the questions about a deprecated API have higher quality score than questions about explicitly referencing deprecation.

Quality over Time

Questions explicitly about deprecation started in 2012 and continue until present day. Overall, these questions do not have much variance in quality, primarily being small numbers about 0. By contrast, questions about the deprecated APIs are dated more recently, likely because the majority of questions collected were about TensorFlow and Keras. Questions about an API post-deprecation have more variance in quality. Overall, the quality for these questions is higher than both questions explicitly about deprecation and general questions about the studied frameworks.



The fanning effect shown in the image shows the assumption that as view count increases, the question score increases.

5.3 Differences in Data

As shown in the graph above, the data has large variance with some outliers in both tails. Therefore, in order to tell if the observed differences are statistically significant, a hypothesis test was run on the data to determine if the quality of is statistically different from each other.

These tests were run to test the following hypotheses:

$$H_0 : The populations of questions about deprecation and not about deprecation have the same mean quality \quad (1)$$

$$H_A : The populations of questions about deprecation and not about deprecation do not have the same mean quality \quad (2)$$

The probability of observing the observed data, given that the questions about deprecation and the overall population of questions have the same mean quality, are outlined in the table below. The tests conducted all state the observed differences in quality are unlikely under the assumptions of the null hypothesis. This provides evidence to suggest that the mean quality of questions explicitly about deprecation is lower than the mean quality of the questions on the studied frameworks and the questions referencing a deprecated library have higher mean quality than questions about the entire population of questions on the studied frameworks.

Hypothesis Test P Values

	Quantile T-Test	Log T-Test	Wilcoxon Rank	Conclusion
Deprecated API & Explicit Reference	0.00	0.00	1.9647e-44	Statistically Different
Deprecated API & All Questions	0.00	0.00	3.7524e-66	Statistically Different
Explicit Reference & All Questions	0.00	1.5805e-256	3.2490e-05	Statistically Different

5.4 Topic Modeling

LDA topic modeling was applied to the data to determine what developers discussed in relation to an API post-deprecation or in relation to deprecation itself. The topic descriptor words were manually investigated and given a name for ease of understanding.

It was found that most topics for questions involving an API post-deprecation were about different specifics of model training and improving accuracy. Another common topic was hardware and using GPUs vs CPU. One topic related to versions and modules. Overall, the deprecation of these APIs were not discussed by developers.

On the other hand, for the questions that explicitly mentioned deprecation, several different topics about deprecation were discussed. The main topics were found to relate to unsupported versions, deprecated models, and the future and deprecation. Similarly to the questions involving deprecated libraries, other topics found outline training, hardware, environments, and miscellaneous errors.

Deprecated API

Topic ID	Topic Name
0	Model Training
1	Hardware Limitations
2	Layer Weights
3	Prediction
4	Training
5	Tensorflow Error
6	LSTM
7	Epochs
8	Environment Error
9	Versions and Batch
10	Convolution
11	Theano Node Error
12	Training Options
13	Model Validation
14	CNN Kernels
15	Shape
16	Hardware
17	NLP
18	Accuracy
19	General Model

Explicit Reference

Topic ID	Topic Name
0	Files
1	Deprecated Models
2	TensorFlow Training
3	Hardware
4	General Coding
5	TensorFlow Deprecation
6	Shell and C
7	Training
8	Environmental Error
9	Code
10	Results
11	Future and Deprecations
12	Theano Installation
13	Build
14	Checkpoint
15	Future and Deprecations
16	Error
17	Training
18	Training Cost
19	DataTypes

Quality of Topics

This was studied by selecting the *"main topic"*, the topic with the most weight to a document, and then assessing the quality of all questions, grouped according to their "main topic". Since the documents were only described by the "main topic", not all topics are included. Weaker topics that contribute less to the discussion are ignored.

Deprecated APIs

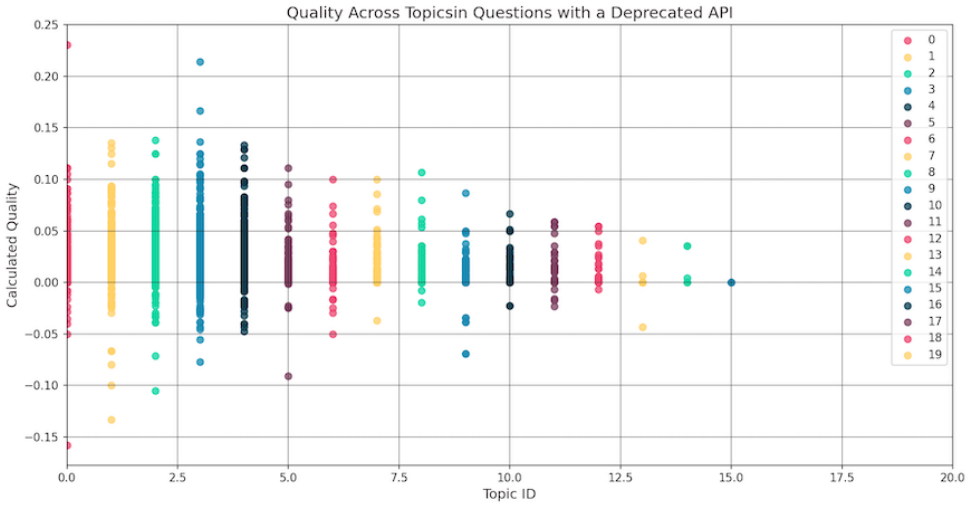
Main Topic ID	Topic Name	Quality
0	Model Training	0.013805
1	Hardware Limitations	0.011523
2	Layer Weights	0.012388
3	Prediction	0.013035
4	Training	0.013460
5	TensorFlow Error	0.012805
6	LSTM	0.008836
7	Epochs	0.013927
8	Environment Error	0.013481
9	Versions and Batch	0.006724
10	Convolution	0.011177
11	Theano Node Error	0.014103
12	Training Options	0.013614
13	Model Validation	0.000532
14	CNN Kernels	0.015126
15	Shape	0.000000

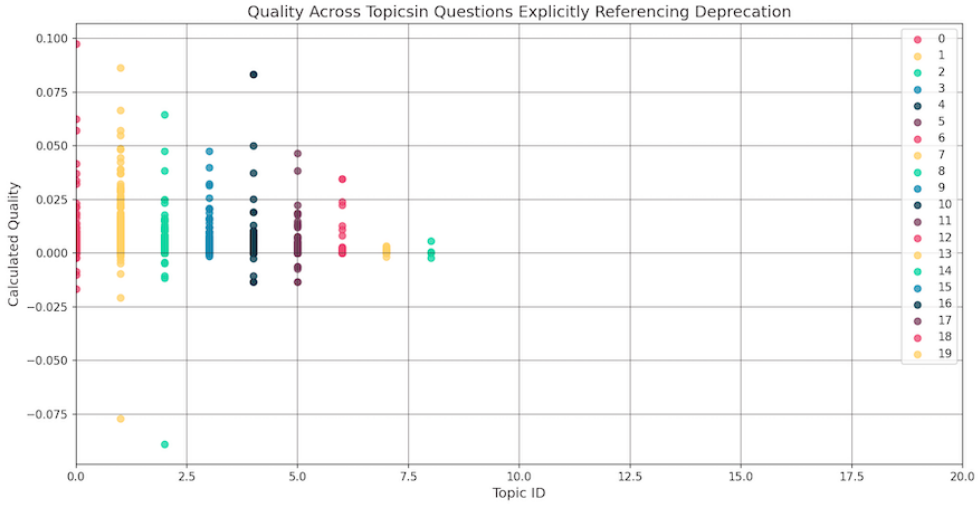
Explicit Reference to Deprecation

Main Topic ID	Topic Name	Quality
0	Files	0.003828
1	Deprecated Models	0.005014
2	TensorFlow	0.002989
3	Hardware	0.004253
4	General Coding	0.004166
5	TensorFlow Deprecation	0.002984
6	Shell and C	0.005147
7	Training	0.000823
8	Code	0.000998

These results show that the questions about model training overall have higher quality than most other topics. However, it is interesting to note, within questions that explicitly reference deprecation of APIs, questions about Model Deprecation had higher quality than questions about model training. Across both datasets, questions with a "main topic" related to deprecation had a lower quality than other questions in the same dataset. See topic 9 in deprecated APIs and topics 2 and 5 in explicit reference to deprecation.

On the other hand, questions explicitly about deprecation are more likely to have a positive quality. Questions about an API after it has been deprecated have higher variance and are more likely to be down-voted. This is illustrated in the graphs below.





5.5 Regression

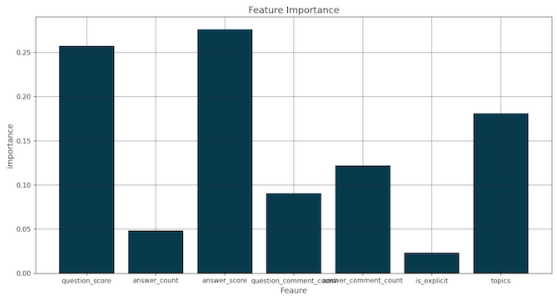
A Random Forest Regression model was trained on the collected data to predict the quality and assess the importance of each predictor variable. The regression model created had an Out-of-Bag Validation of *0.7568900018162473*. Since this score is relatively low, the results of the regression model should be taken with the performance of the model in consideration.

The most important features are the scores of the question and the answer. This is expected, as the subjective community scores are used in the calculation of the quality. However, the third most important feature is the main topic discussed. To compare, an explicit reference to deprecation is the least important determinant of quality. This

result suggests that while the topic of deprecation is a factor in assessing the quality, the quality questions with an explicit reference to deprecation but are not primarily about the deprecation is more dependent on the main topic than the reference.

Feature Importance

Feature	Importance
Question Score	0.25752
Answer Score	0.27639
Answer Count	0.04878
Question Comment Count	0.09051
Answer Comment Count	0.12206
Explicit Reference	0.02354
Main Topic	0.18121



6 HUMAN EVALUATION

Since there were very few questions that explicitly referenced deprecation within the dataset, a manual investigation was conducted on the questions about an API post-deprecatiom. A total of 366 with 218 answers were randomly sampled for investigation.

	Asked About Deprecation	Answer About Deprecation	Total Deprecation
Number of Questions:	17	16	33
Percentage of Questions:	4.6448	4.3715	9.0164

It was found that not very many questions are asked about the deprecation of an API. For most users, the deprecation was ignored. Overall, a large number of questions seemed largely unfocused, with the questioner not knowing what is wrong with their code. In the cases where an issue due to deprecation was discussed, it was just as likely to be pointed out by the user answering the question than the user asking the question. The most popular version issues were Keras's recent deprecation in version 2.4.0, which discontinued multi-backend Keras in favour of the TensorFlow implementation, or about Theano's deprecation. Otherwise, many questions were about batch size, data shapes, hardware specifications, improving accuracy, or one-hot-encoding.

One trend was found that is important to note, as it threatens the validity of the analysis. Many users would mention a framework or API, despite the question not specifically being about the framework or API. Therefore, since some API names are duplicated in two different frameworks, while only one is deprecated, a question is incorrectly filtered as including a deprecated framework.

7 RELATED WORKS

There have been several different proposed metrics for quality. While this study has used the results of Ravi et al[9], the related works outline other metrics. Baltadzhieva and Chrupala survey how question quality affects answer rate [1]. Tian *et al.* studies answer acceptance[15]. Shah studies how to determine the best answer to a question[12]. Fu *et al.* combines textual and non-textual features to determine quality[3]. Zhang *et al.* studies the problematic effects of obsolete answers[20]. Other related works study different important elements on Stack Overflow. Ren *et al.* studies how controversial an answer is[10]. Diyanati's findings determine no significant relationship between community scores and

the expertise level of users, instead focusing on comment mining[2]. Wang *et al.* propose an answer recommendation system using expertise level, topical interest, and activeness[17]. Zhang *et al.* notes that answers should be supplemented by comments[19]. Zhou *et al.* investigate the methods provided by Stack Overflow, bounties, to increase answer rate[21]. Studies into Stack Overflow and other forums are important because they are useful tools for many developers.

Studying deprecation and its effects is also important. Wang *et al.* notes how API deprecation is not handled well among developers[16]. Yasmine *et al.* notes that, for REST APIs, most do not follow deprecated-removed protocol[18]. Similarly, Sawart finds that most consumers do not react to deprecated features, while recommended replacement is rarely implemented[11]. Automatic updates to code bases after a deprecation have been proposed. Thung *et al.* propose automated updates to Android APIs, noting the examples that are not updates[14]. Nishi and Damevski propose automatic identification of valid APIs on online tutorials[8]. From these findings, there is evidence that developers do not react ideally to deprecation and prefer to ignore it entirely. Lui *et al.* notes the importance of compatibility debt in debt for deep learning systems, especially with Keras[6]. Compatibility is a possible explanation for developer's reaction to deprecation.

8 THREATS TO VALIDITY

Human Evaluation: Human evaluation was used to interpret the topics determined via LDA. The analysis of how the topics discussed affect the quality of a question and answer is dependent on properly understanding the overall topic. While most topic names came directly from the keywords of the topic, it is possible that some keywords were misunderstood while interpreting the topics.

Quality Metric: Any analysis of quality is dependent on the metric used to quantify quality. The metric used to calculate the quality of a question and answer pair is dependent on three assumptions. First, the view count of a question and an answer are similar. Since questions are most likely to be answered within two days, this number is assumed to be similar. Second, as view count increases, the score increases proportionately. Finally, the view count of a question about deprecation is assumed to decrease over time, as more versions are released since the deprecation.

Regression Score: The model created had an Out-Of-Bag score of 0.7569. This score is low. Since the metric for quality was created by a non-linear combination of other columns, some of these columns needed to be removed. When all columns were added as parameters, the model preformed well only using these parameters, even when limiting the number of features used as a split. The results found in the regression tree must be considered with the removed parameters and low performance in mind.

9 CONCLUSION

In this study, it was found that questions about a deep learning API after its deprecation, specifically training models using said API, were considered higher quality than questions specifically about deprecation. More importantly, questions asked about an API after its deprecation were calculated to have a higher average quality than the average quality questions about the studied frameworks. This suggests that deprecation of an API tends to overlap with a higher question quality. By contrast, questions primarily about deprecation, by explicitly referencing deprecation or primary topic being deprecation, have a lower average quality than questions about the studied frameworks.

This result is conflicting. However, it shows that deprecation affects developers in many ways. As the related works state, most developers ignore deprecation and do not adopt to the most recent version. This reflects in the metrics of quality. Since developers do not replace the API after it has been deprecated, questions about these APIs are of higher value to the community. By contrast, questions about deprecation are rated of lower value to the community because it

is more rare that developers replace the API. As shown by the results of this study, deprecation leads to an increase in quality of answers on Stack Overflow.

Future studies include the following question: "Do the users discuss the backward compatibility issues"? The reference that inspired this question is inspired by *An Empirical Analysis of Backward Compatibility in Machine Learning Systems*[13].

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